
3.6.3 Data should be analysed by means of an appropriate statistical test. This allows calculation of the probability of an event being due to chance. Appropriate conclusions should be drawn and scientific knowledge from the A Level specification should be used to explain these conclusions.

Candidates should be able to

- select and justify the choice of an appropriate statistical test from the following:
 - standard error and 95% confidence limits
 - Spearman rank correlation
 - χ^2
- construct an appropriate null hypothesis
- calculate the test statistic given a standard scientific calculator
- interpret the calculated test statistic in terms of the appropriate critical value at the 5% significance level, making reference to chance, probability and acceptance or rejection of the null hypothesis
- draw valid conclusions, relating explanations to specific aspects of the data collected and applying biological knowledge and understanding from the A Level specification.

Practical work carried out in the context of Units 4 and 5 should enable candidates to gain experience of

- selecting, using and interpreting an appropriate statistical test from the following:
 - standard error and 95% confidence limits
 - Spearman rank correlation
 - χ^2